

Thm. Let f be an analytic on the closed disk $|z - z_0| \leq r$. Then

$$f(z_0) = \frac{1}{2\pi} \cdot \int_0^{2\pi} f(z_0 + r \cdot e^{i\theta}) d\theta.$$

Proof. By the Cauchy's integral formula,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_{|z - z_0| = r} \frac{f(z)}{z - z_0} dz$$

Then, for a parameterization $z(\theta) = z_0 + r \cdot e^{i\theta}$ ($0 \leq \theta \leq 2\pi$),

$$\begin{aligned} &= \frac{1}{2\pi i} \cdot \int_0^{2\pi} \frac{f(z_0 + r e^{i\theta})}{r e^{i\theta}} \cdot r \cdot i e^{i\theta} d\theta \\ &= \int_0^{2\pi} f(z_0 + r e^{i\theta}) d\theta \quad \square \end{aligned}$$

This theorem is called the Gauss's Mean-Value Theorem.

Maximum Modulus Theorem.

If non-constant f is analytic in a domain D , then $|f|$ cannot attain a maximum in D .

Proof. Suppose that $|f|$ attains maximum at $z_0 \in D$. (Then f must be a constant).

Since D is open, for some $r > 0$, $\overline{B_r(z_0)} \subset D$. Then, the Gauss mean-value Thm gives

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + r e^{i\theta})| d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0)| d\theta = |f(z_0)|$$

$|f(z_0)|^*$ is maximum.

So that $|f(z_0)| = \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + r e^{i\theta})| d\theta$, or

$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0)| - |f(z_0 + r e^{i\theta})| d\theta = 0.$$

By the assumption, the term in the integrand is non-negative, thus

$$|f(z_0 + r e^{i\theta})| = |f(z_0)| \quad \text{for all } 0 \leq \theta \leq 2\pi.$$

for all $0 < \epsilon \leq r$, $|f(z_0)| = |f(z_0 + \epsilon e^{i\theta})|$ and so $|f(z)| = |f(z_0)|$ for all $z \in \overline{B_r(z_0)}$.

So, f is constant, the identity theorem gives the result.